

Lab1: C Basics

Data Science

Task 1: Sum of Digits

- Write a C program to input a natural number and calculate sum of digits using for loop.
 - Input: 1234 Output: 10
 - Input: 98 Output: 17

```
Enter any natural number: 1234
Sum of digits = 10
```

```
Enter any natural number: 98
Sum of digits = 17
```

- Hint: The main idea to find sum of digits can be divided in three steps.
 1. Extract last digit of the given number.
 2. Add the extracted last digit to sum.
 3. Remove last digit from given number. As it is processed and not required any more.

Task 2: Find largest number

- Write a C program to ask the user input three numbers (separate by white space) and output the largest of three input numbers.
- Input: 1 2 3 Output: 3
- Input: 12 45 21 Output: 45

```
Input the values of three numbers : 1 2 3
The largest number is 3.
```

```
Input the values of three numbers : 12 45 21
The largest number is 45.
```

Task 3: String & Array

- Write a program to read a string using “%s” in a scanf statement and separate individual characters from the string and output the individual characters (separated by white space)
 - You can assume that total number of characters will not exceed 50
 - Input: hello Output: h e l l o
 - Input: world Output: w o r l d

```
Input the string : hello
The characters of the string are :
h e l l o
```

```
Input the string : world
The characters of the string are :
w o r l d
```

- Hint: Loop to print each individual character of the string until the null terminator '\0' is encountered

Task 4: Count words in a sentence

- Write a program to read a sentence and count the number of words in the input sentence.
 - You can assume that total number of characters of the input sentence will not exceed 50
 - Input: hello world Output: 2
 - Input: i love data science Output: 4

```
Input the string :  
hello world  
The No. of words in the given string is: 2
```

```
Input the string :  
i love data science  
The No. of words in the given string is: 4
```

- Hint:
 - Use `fgets()` to read the input sentence
 - Loop until the end of the string (`'\0'`)
 - Check whether **the current character is white space**;
 - If it is, then check if the following character is **neither whitespace, a newline, nor a tab**. If that's the case, **increment the count by one**.
 - Final **count add by one** is the number of words.

Task5: Recursion - Factorial

- Write a program in C to read in a natural number and calculate the factorial of the number using **recursion**
 - $\text{Fac}(n) = n * (n-1) * \dots * 1$, for all $n > 0$
 - Input: 5 Output: 120
 - Input: 1 Output: 1
- You may build your recursion as follows:
 - **Base**, if $n=1$: $\text{Fac}(1)=1$
 - **Step**, if $n>1$, $\text{Fac}(n) = n * \text{Fac}(n-1)$
- Note that your program should contain
 - a main function, which does IO
 - and a recursive function, **int fac(int n)**, which computes the factorial

```
Input  a number : 5
The Factorial of 5 is : 120
```

```
Input  a number : 1
The Factorial of 1 is : 1
```

Task 6: Recursion - Reverse String

- Read in and **reverse a string** using **recursion**
- Note that your program should contain
 - a main function, which does IO
 - and a recursive function as follows, which reverses the string
- **Note: You may assume that the string size is less than 50.**

```
void reverseR(char* str, int start, int end){  
    // str is the string  
    // start and end are the start and end  
    // indices of the substring that needs to be reversed.  
}
```

Task 6: Recursion - Reverse String

- You may build your recursion as follows.
 - **Base**, if the string size is 1 or 0: nothing is done.
 - **Step**, if the string size is at least 2: swap the first and last chars and **reverse the remainder using recursion**.
- Input: hello Output: olleh
- Input: hi Output: ih
- Note that your program should contain
 - a main function, which does IO
 - and a recursive function as follows, which reverses the string

```
Please enter a String:hello
The reversed string is: olleh
```

```
Please enter a String:hi
The reversed string is: ih
```

Submission

- Save your .cpp files as `t1.c`, `t2.c...`, submit the files to iSpace.